

General Fourier Formula

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[\left(a_n \cos\left(\frac{n\pi}{L}x\right) \right) + \left(b_n \sin\left(\frac{n\pi}{L}x\right) \right) \right]$$

y-offset of the shape
[avg value]

Adjust the shape

Half period

Actual wave

Test wave

Actual wave

Test wave

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

Avg value

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

Resonance

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Resonance

When $L = \pi$, we get the following...

The Fourier Series of a 2π -periodic function $f(x)$ can be written as

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

y-offset of the shape

Adjust the shape

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

This literally gives the avg value of $f(x)$ [Total Area / 2π]

Finds the "vertical centre" of the shape

Actual wave

Test wave

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

scaling factor

Resonance

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

Actual wave

Test wave

scaling factor

Resonance

This new "resonance" function will only have a non-zero area when the waves sync up so that even their -ve areas become +ve, because $(-ve \times -ve = +ve)$. If the waves don't sync, it will create equal amounts of +ve & -ve areas that perfectly cancel out to zero

Example

Determine the Fourier Series for

$$f(x) = x+1, -2 < x < 2, f(x) = f(x+4)$$

$$2L = 4 \Rightarrow L = 2$$

$$a_0 = \frac{1}{2(2)} \int_{-2}^2 (x+1) dx$$

$$= 1 \#$$

$$a_n = \frac{1}{2} \int_{-2}^2 (x+1) \cos\left(\frac{n\pi}{2}x\right) dx$$

$$\vdots$$

$$= 0$$

$$b_n = \frac{1}{2} \int_{-2}^2 (x+1) \sin\left(\frac{n\pi}{2}x\right) dx$$

$$\vdots$$

$$= -\frac{4}{n\pi} \cos(n\pi)$$

$$= -\frac{4}{n\pi} (-1)^n$$

$$= \frac{4}{n\pi} (-1)^{n+1}$$

$$\therefore \text{Fourier Series} = 1 + \sum_{n=1}^{\infty} \left[\frac{4}{n\pi} (-1)^{n+1} \sin\left(\frac{n\pi}{2}x\right) \right] \#$$

Partial Sum, S_N

[Work in Progress Signal]

Since we can't realistically sum up infinite no. of trigo terms, we use partial sums.

Partial sum answers the question...

"If I am forced to stop at a specific number of trigos, what is the absolute best reconstruction of the original signal I can get"

$$S_N(x) = a_0 + \sum_{n=1}^N \left[a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right) \right], \text{ for finite } N$$

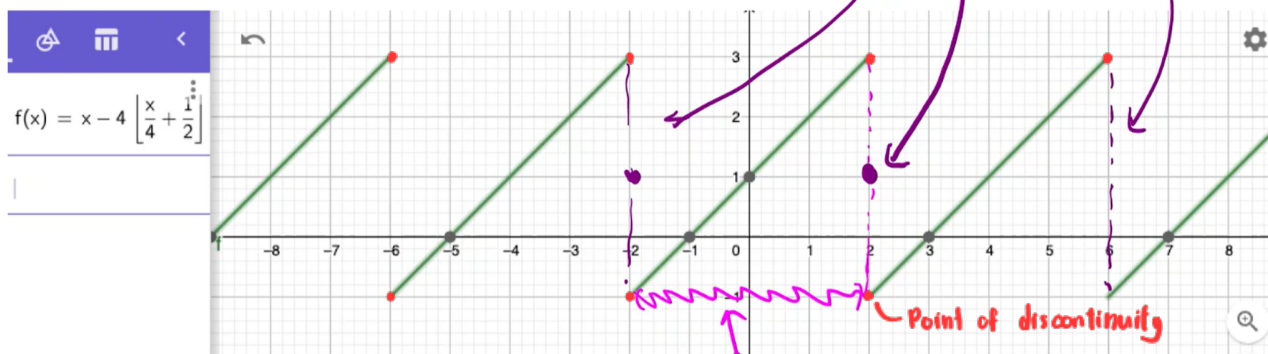
Do note that S_N is made up of valid functions [sin & cos], its resultant function will also be a valid function [No vertical jumps (1 input $\Rightarrow$ >1 output)]
 $\therefore$ If $f(x)$ have points of discontinuity, x_d , the Fourier Series will converge to the avg of LH & RH limits, that is

$$\frac{1}{2} [f(x_d - \delta x) + f(x_d + \delta x)]$$

Example 4.2.2

[Avg of LH & RH Limits]

Fourier series converges to 1 here



Fourier series converges to $f(x)$ for $-2 < x < 2$
[For $N \rightarrow \infty$, Fourier series converges perfectly to $f(x)$]

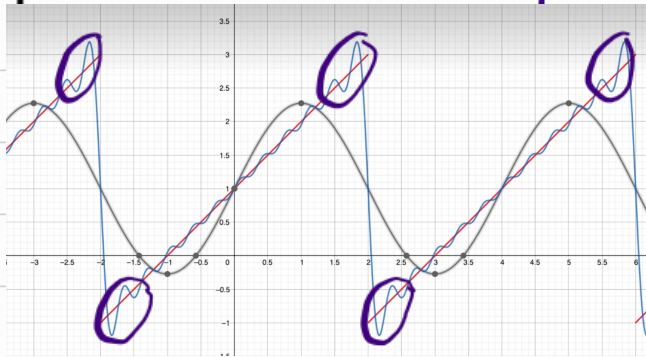
Half Range Expansion Approx.

Mathematical trick to "force" a non-repeating function to work w Fourier Series. We essentially just expand/extend the original function [without changing the original func. within that interval] however we want.

$\therefore$ To make life easier, we expand it to either be $b_n = 0$ perfectly even or $a_n = 0$ perfectly odd.

For the best Fourier Series approx., these are the priorities

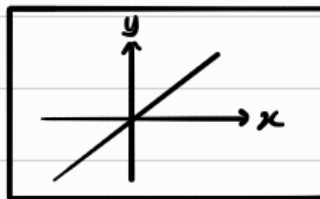
1) Avoid jump discontinuities [Avoid Gibbs phenomenon]



2) Minimise kinks [They reduce the rate of convergence]

Example

Consider $f(x) = x$, for $0 < x < 1$



Copy & Paste

[aka. "standard/full range expansion"]



Have jump discontinuity

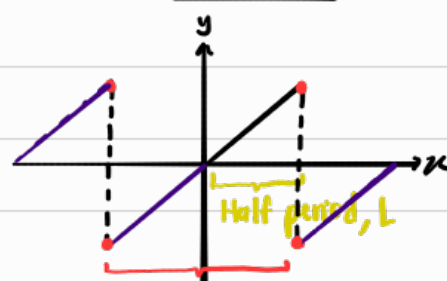
Half - Range Expansion

Even



Don't have jump discontinuity

Odd



Have jump discontinuity

$\therefore$ Even will be used to approx $f(x)$ w Fourier series.

Half-range expansion essentially just gives us the freedom to engineer the boundaries for a better approx w Fourier series

